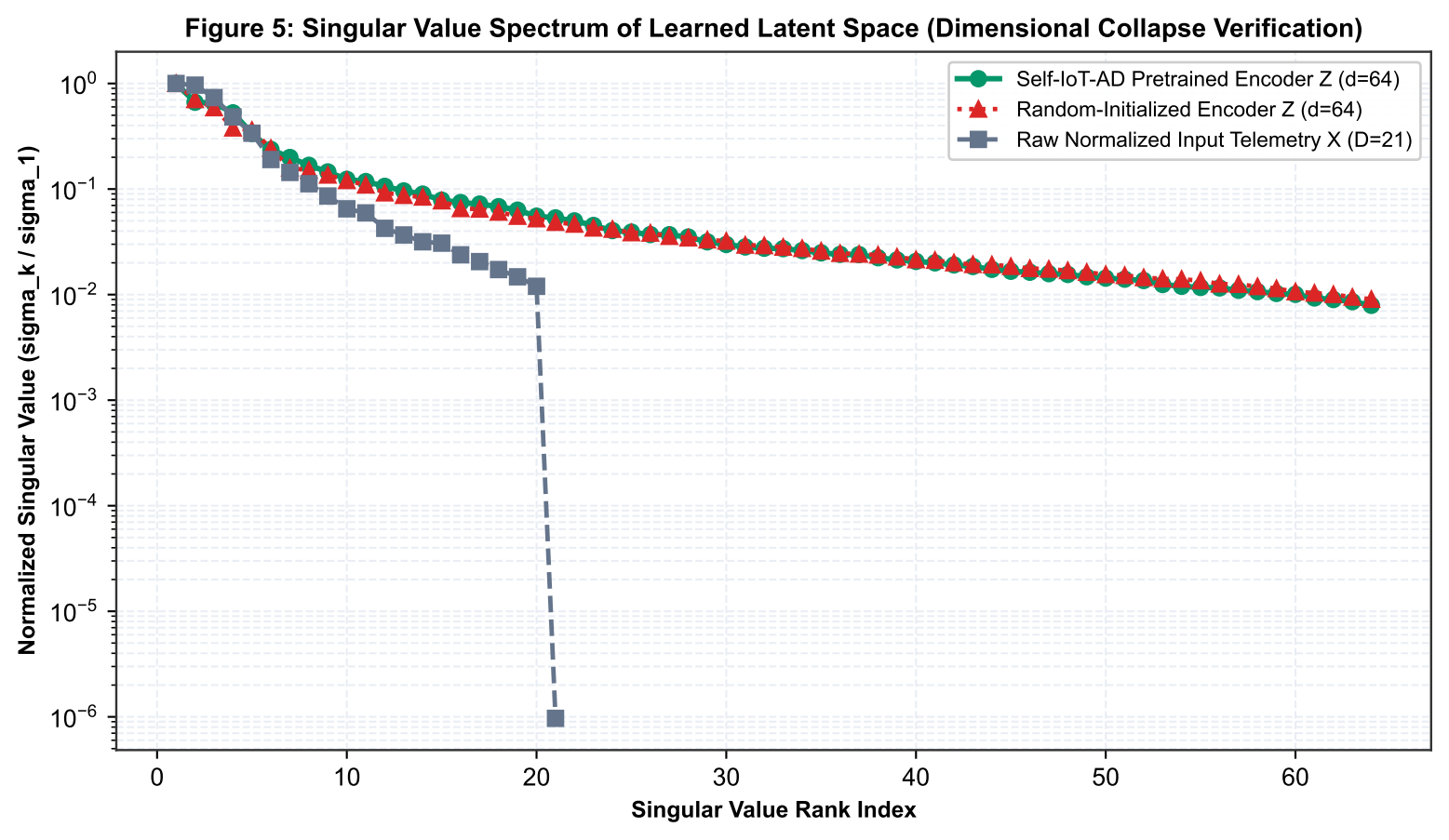
Figure 5: Singular Value Spectrum of Learned Latent Space (Dimensional Collapse Verification)
Self-IoT-AD Pretrained Encoder Z (d=64)
Random-Initialized Encoder Z (d=64)
Raw Normalized Input Telemetry X (D=21)
Normalized Singular Value (sigma_k / sigma_1)
Singular Value Rank Index
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
0
10
20
30
40
50
60